

Kuldeep Lodhi

SENIOR SOFTWARE ENGINEER

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Summary

Senior Software Engineer with 7+ years building high-performance, data-intensive platforms for large-scale logistics, retail, and inventory operations. Specialized in React, TypeScript, UI/UX-focused frontend architecture, and real-time systems (Kafka, SSE), with hands-on experience integrating **Google Maps SDK** for GPS tracking, route visualization, and facility mapping. Proven **Team Lead** experience mentoring engineers and driving Agile delivery, with a track record of shipping enterprise platforms serving 1,300+ stores, 6,000+ daily users, and INR 50 Cr in managed inventory.

Professional Experience

Reliance Jio Platforms

Jun 2024 -- Present

Software Development Engineer – 2

Navi Mumbai, India

- Led frontend development for an enterprise inventory platform serving 1,300 stores across 6 brands, managing INR 50 Crores in inventory with 6,000 daily active users
- Built 50+ reusable TypeScript components and standardized Redux architecture, reducing frontend duplication by 30%
- Improved dashboard performance by 45% using memoization and virtualization, enabling real-time operational dashboards
- Acted as **Team Lead** for a 4-member frontend team, driving UI/UX consistency, code reviews, and Agile delivery
- Implemented Docker containerization and CI/CD pipelines via Azure DevOps, enabling frequent production releases

Mobile Programming (Client: Reliance Jio)

Dec 2023 -- Jun 2024

Software Developer

Mumbai, India

- Worked as a client-side developer deployed at Reliance Jio, contributing to the early development of an enterprise inventory management platform
- Collaborated directly with Jio's internal engineering team on frontend modules using React and TypeScript, aligning with existing architecture and coding standards
- Contributed to initial feature development, system integrations, and data handling workflows for large-scale retail operations

Algonauts Advisory

Nov 2021 -- Dec 2023

Senior Software Developer

Mumbai, India

- Architected a full-stack trading simulator platform using Svelte.js serving 400 daily users and processing 8K trades/day
- Collaborated directly with founders to design product architecture and trading workflows
- Implemented an ML-based portfolio recommendation system, increasing engagement by 25%
- Scaled the engineering team from 1 to 3 developers while maintaining 99.5% uptime

Ugam Solutions

Jun 2019 -- Nov 2021

Senior Analyst

Thane, India

- Delivered 15 enterprise analytics projects for CDC, University of Chicago, Google Screenwise, and IPSOS
- Developed JavaScript libraries and dashboards, reducing report generation time by 40%

Key Projects

AnyTrac RFID Platform

React, TypeScript, Redux, Node.js, Kafka, Google Maps SDK

- Implemented unified RFID-based inventory tracking, enabling 50-item carton scanning in 1 second vs. 5 minutes manually
- Integrated **Google Maps SDK** to power real-time GPS location tracking of in-transit inventory and facility-level floor-plan mapping for zone-based stock visibility
- Built route visualization on top of the Maps SDK to plot and monitor inventory movement paths across store clusters, improving in-transit visibility for operations teams
- Implemented a **Tag Management System (TMS)** to centrally deploy and manage marketing/analytics tags across store portals, reducing manual code changes for tracking updates
- Improved inventory accuracy from 75% to 98% and reduced shrinkage by 40%
- Designed an event-driven architecture where Product SOH (Stock on Hand) and tracking updates are pushed to Kafka queues and consumed by downstream services
- Supported Store Portal, Cluster Portal, and Self-Checkout systems across 1,000 retail locations

JVA Retail Video Analytics Platform

React, TypeScript, Konva, Kafka, Real-time Streaming

- Built dashboards for an AI-powered video analytics platform converting camera streams into retail operational insights
- Implemented modules for traffic analytics, queue monitoring, customer demographics, and store heatmaps
- Developed a store planner interface for floor-plan mapping and camera placement, enabling zone-based visitor analytics
- Integrated a Kafka streaming pipeline where camera feeds are published to Kafka and consumed by multiple analytics pipelines processing feeds against defined Regions of Interest (ROIs)
- Implemented Server-Sent Events (SSE) enabling real-time alerts for queue thresholds, congestion, and system events

Reliance Jewels Inventory Platform

React, TypeScript, Redux

- Built an enterprise inventory management platform for Reliance Jewels stores, supporting daily stock reconciliation and store operations
- Developed dashboards for Stock-on-Hand (SOH), inward/outward tracking, and real-time stock visibility across store clusters
- Enabled store managers to monitor inventory movements, reducing reconciliation time and improving operational accuracy
- Implemented performance optimizations and reusable UI components, improving dashboard responsiveness for large datasets

Reliance Foundation Scholarship Portal

React, Node.js, MongoDB

- Built a scholarship management system handling 50,000 annual applications with OCR-based document ingestion
- Migrated a legacy K2 workflow system to a MERN architecture, improving scalability by 200%
- Developed an interview management portal supporting 50 concurrent interview sessions with automated evaluation workflows

Technical Skills

Frontend: React, TypeScript, Next.js, Redux, Svelte, Konva, Chart.js, ApexCharts

Maps & Geospatial: Google Maps SDK/API, Real-time GPS Tracking, Route Visualization, Geofencing, Floor-plan/Facility Mapping

UI/UX: UI Component Design, UI/UX Consistency, Material-UI, Tailwind CSS, Bootstrap, Responsive Design

Backend: Node.js, Express, Python, Django, REST APIs, Microservices

Streaming & Messaging: Apache Kafka, Event-driven Architecture, Tag Management System (TMS)

Database: MongoDB, PostgreSQL, MySQL, Redis

DevOps & Cloud: Docker, CI/CD, Azure DevOps, AWS, Git

Performance Optimization: Memoization, Bundle Optimization, Code Splitting, Lazy Loading, Component Design Patterns

System Design: Scalable Systems, Distributed Systems

Leadership & Collaboration: Agile/Scrum (Sprint Planning, Story Estimation, Standups, Retrospectives), Team Lead (4 Engineers), Code Reviews (30/Sprint), Cross-functional Collaboration, Stakeholder Management

Education

Bachelor of Engineering in Computer Engineering

Vidyalankar Institute of Technology, Mumbai

2015 – 2019

CGPI: 7.95/10